GlucoTwin

Digital Twin Challenge 2026 | Two-hour synthetic glucose forecast

A reproducible digital twin proof of concept

Team: [Add team member names] College: [Add institution]

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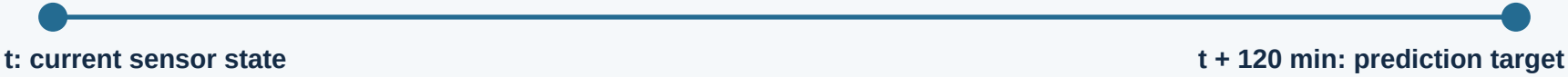
Synthetic data only. Not for clinical use.

The prediction problem

Can the current virtual patient state inform a two-hour forecast?

- People with Type 2 diabetes need ongoing glucose monitoring. This prototype explores a forecasting workflow, not a clinical intervention.
- The defined target is sensor glucose in mg/dL exactly 120 minutes after the current timestamp.
- At a 15-minute cadence, the label is the eighth subsequent sensor reading.
- The dashboard makes the prediction horizon visible beside the current simulated reading.



Healthcare use case

A research-facing interface concept for review of trends

- A synthetic patient profile supplies age, BMI, HbA1c, diabetes duration, and a medication category.
- A simulated sensor stream supplies glucose, heart rate, activity, sleep flag, and time-of-day features.
- A reviewer can change patient, advance the virtual time, and compare a forecast with the current state.
- A future governed study could evaluate whether such forecasts add value to a care workflow. This demo does not test that clinical question.

The gap this prototype addresses

A reproducible engineering workflow, with its evidence boundary made explicit

Prototype controls

- A stated two-hour label
- Time-ordered evaluation
- Current-state prediction inputs
- Visible synthetic / prediction labels

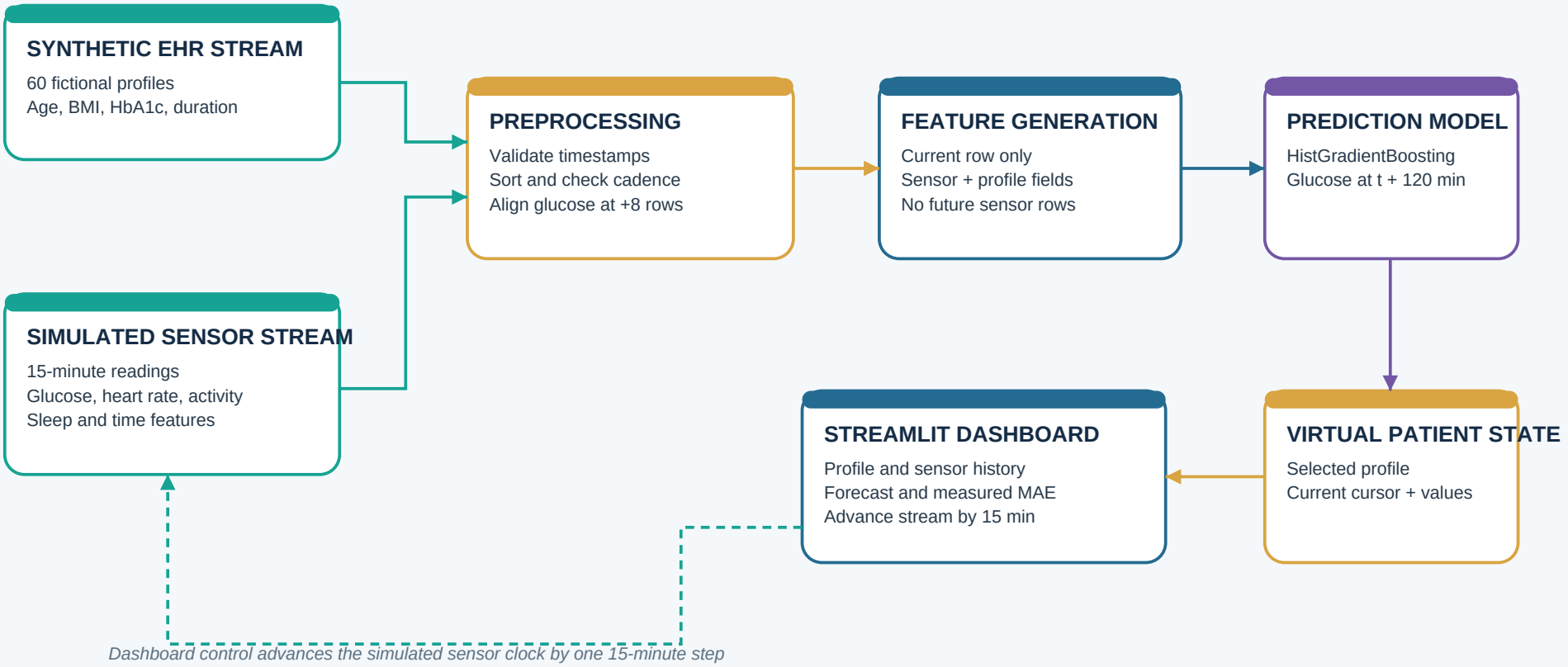
Failure mode avoided

- Target horizon can be ambiguous
- Random splits can hide time leakage
- Forecasts can accidentally see later rows
- Simulated values can be mistaken for observations

This describes the engineering gap addressed by this submission. It is not a comparison with named commercial or clinical systems.

Architecture

Two local synthetic streams feed a time-aligned forecast



Synthetic data

Seeded and local. No real EHR or device data are used.

PROFILES

60

SENSOR SAMPLES

120,960

CADENCE

15 min

COVERAGE

21 days

PATIENT RECORDS

Synthetic EHR

SENSOR FIELDS

5 + time

Profile fields include age, sex, BMI, HbA1c, duration, and medication category. Sensor fields include glucose, heart rate, activity, sleep, and cyclic time features.

Target, features and leakage controls

The model sees the present row, while evaluation respects time

- Label: glucose at t + 120 minutes, aligned by shifting eight records within each patient stream.
- Input features: current glucose, heart rate, activity, sleep, time-of-day, age, BMI, HbA1c, and diabetes duration.
- The inference helper selects only those columns from the selected current row and patient profile.
- The chronological holdout reserves later timestamps for testing and purges training rows whose target reaches the test cutoff.

No future sensor rows are passed to the inference feature builder.

Model and baseline

A local scikit-learn regressor compared with persistence

HistGradientBoostingRegressor

HistGradientBoostingRegressor fitted on the earlier portion of the pooled timeline. The saved artifact contains the model, feature names, and evaluation metrics.

Baseline

Predict the two-hour glucose target using the current glucose value unchanged. This simple persistence baseline is evaluated on the same held-out rows.

No hyperparameter search or clinical calibration is claimed.

Measured evaluation

One reproducible synthetic run. No clinical validation.

MODEL MAE

3.335 mg/dL

PERSISTENCE MAE

8.580 mg/dL

Chronological split

- Train: 95,880 rows
- Test: 24,120 later rows
- Cutoff: 2026-10-15T17:30:00+00:00
- 120-minute label purge before test boundary

Synthetic-data MAE is not evidence of real-world accuracy, safety, benefit, or generalization.

Dashboard demonstration

Actual Streamlit AppTest state transitions, not a mock screenshot

Interaction	Current glucose	Prediction at +2h	Stream cursor
SYN-001 initial	95 mg/dL	95.0 mg/dL	00:00 UTC
Advance 15 min	97 mg/dL	95.2 mg/dL	00:15 UTC
Select SYN-002	129 mg/dL	131.2 mg/dL	00:00 UTC

Verified: app renders without exceptions; selecting and advancing the stream recomputes the forecast.

A browser screenshot was unavailable in this run, so no screenshot or UI mock is presented.

Limits, next steps and submission status

A prototype for research discussion, not a care tool

- All profiles, sensor readings, and metrics are generated. The simulator is simplified and unvalidated.
- No prospective, external, subgroup, safety, fairness, or clinical validation has been performed.
- Next: confirm challenge protocol, obtain appropriate governance and consent, validate on suitable data, and assess subgroup performance before any clinical research claim.
- Demo video: pending real recording and upload. Team and institution details remain placeholders.

MIT licensed | Full setup, model results, and limitations are in README.md